Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimensional Political Opinion State Space

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Abstract

Information operations exploit heterogeneity in individual susceptibility to shift political opinions at scale, yet the joint profile of which psychological and demographic characteristics amplify vulnerability—across diverse attack mechanisms and opinion domains—remains poorly characterised. We present a multi-agent simulation pipeline in which 80 synthetic pseudoprofiles, parameterised along 85 feature dimensions spanning Big Five personality, age, sex, a Dual Process Inventory, a Digital Literacy Inventory and a Political Engagement Inventory, are crossed with six cyber-manipulation attack vectors (narrative framing, fear appeal, astroturfing, pseudo-expert authority, deepfake audio impersonation, LLM-personalised chatbot persuasion) and eight political-opinion leaves spanning four top-level domains—Defence and National Security, Foreign Policy and Geopolitics, Immigration and Citizenship, and Information Integrity and Platforms (N = 3,840 attacked scenarios). The primary outcome, adversarial effectivity (AE), captures signed directional opinion movement toward an operator-specified goal. Intraclass correlation partitions variance into between-profile and attack–opinion context components; both ICC(1) for absolute displacement and adversarial effectivity collapsed to ≈ 0 in the broader multi-domain design. A conditional ridge regression susceptibility index with leave-one-group-out hierarchical importance decomposition isolates moderating personality dimensions; block-bootstrap confidence intervals ($B = 600$) quantify uncertainty in profile-level coefficients. The present study confirms that attack–opinion context dominates profile-level variance: the multivariate profile-level OLS model is non-significant as a whole ($R^2 = 0.151$, Adj. $R^2 = 0.055$; $F(8, 71) = 1.612$, $p = .137$), no predictor survives Benjamini–Hochberg FDR correction, and mean AE is near zero ($\overline{\text{AE}} = -0.84$; 48.2% of scenarios producing positive AE). Extraversion is the only predictor whose bootstrap 95% CI excludes zero ($\hat{\beta} = +1.61$, $p = .023$; 95% CI $[0.47, 3.00]$), a nominally susceptibility-amplifying association that does not replicate the Conscientiousness signal observed in the single-domain predecessor study (Defence-only, $N = 100$). The pipeline provides open, reproducible infrastructure for defender-oriented susceptibility auditing and highlights the fragility of single-domain personality–susceptibility claims.

Introduction

Political influence operations increasingly exploit the gap between information volume and cognitive processing capacity. Rather than relying on technical intrusion, they operate through selective framing, fabricated social consensus, pseudo-expert endorsement, and identity-congruent narrative design—mechanisms that shape the epistemic conditions of political judgement rather than changing any specific belief (Bennett and Livingston, 2018; Lazer et al., 2018; Lewandowsky et al., 2017). The question this raises is not simply whether manipulative content is detectable, but whether adversarial exposure predictably shifts opinions in operationally useful directions, and whether that susceptibility varies systematically across individuals. Coordinated disinformation campaigns have grown in both scale and sophistication, with organised social media manipulation now documented across dozens of countries (Bradshaw and Howard, 2019), amplifying the urgency of understanding which population segments are most vulnerable and why.

Recent policy and academic discourse has framed this vulnerability under the term *cognitive warfare*: the deliberate targeting of human attention, trust, and decision-making as operationally relevant terrain (Hung and Hung, 2022; Johns Hopkins University Applied Physics Laboratory and NATO Allied Command Transformation, 2021; Paulauskas, 2024). We adopt *cognitive sovereignty*—the capacity for self-directed political belief formation under adversarial conditions—as the substantive security construct. Susceptibility is then operationalised as the degree to which post-exposure opinions move in directions intended by a hypothetical adversarial operator, not as simple opinion change magnitude. This framing positions the pipeline as a *defender's tool*: identifying population segments for which particular attack–opinion combinations produce the largest effective shifts enables targeted inoculation, media literacy interventions, and institutional protective measures (Kozyreva et al., 2020; Roozenbeek et al., 2022).

The emergence of generative AI substantially changes the threat surface. Large language models (LLMs) enable personalised manipulation at scale (Goldstein et al., 2023): deepfake audio impersonating authority figures, AI-generated images fabricating events, and personalised persuasion bots calibrated to individual psychological profiles are no longer theoretical. Evidence that persuasive messaging tailored to recipients' Big Five personality traits can substantially increase compliance and engagement (Hirsh et al., 2012) suggests that adversaries equipped with psychographic data may exploit these individual differences to maximise persuasive impact. Understanding which profile characteristics moderate susceptibility to these novel attack types—alongside traditional narrative framing and coordinated amplification—is therefore an urgent scientific and security priority.

The methodological approach grounds itself in demonstrations that LLM agents produce socially and demographically conditioned behaviour approximating human response distributions at aggregate levels (Argyle et al., 2023; Park et al., 2023). These findings motivate using LLM-generated synthetic profiles as tractable proxies for studying how personality and demographic dimensions interact with adversarial exposure, with results interpreted as properties of the simulation rather than direct claims about human populations. A critical design choice is *why simulation rather than direct elicitation*: prompting an LLM to rate

a profile's vulnerability yields an opaque scalar judgment. The pipeline instead produces a structured, conditional measurement system: task-specific susceptibility estimates across a factorial attack–opinion design, repeated indicators suitable for structural equation modelling (SEM), and an auditable four-stage pipeline of baseline elicitation, exposure generation, post-exposure assessment, and realism audit. This object can be compared to human experimental data through criterion validity, convergent validity, and stability testing in ways a scalar cannot.

The present study addresses the question: *Which inter-individual differences—in personality and demographics—moderate adversarial effectivity across diverse attack mechanisms and political-opinion domains?*

We report a factorial simulation study: 80 pseudoprofiles parameterised across 85 feature dimensions (Big Five personality and facets; Dual Process Inventory, Digital Literacy Inventory, and Political Engagement Inventory; age; sex) crossed with six attack vectors—including two AI-driven synthetic-media attacks (Deepfake Audio Speech Impersonation, LLM Chatbot Personalised Persuasion)—and eight opinion leaves spanning four top-level domains (Defence and National Security; Foreign Policy and Geopolitics; Immigration and Citizenship; Information Integrity and Platforms), for $N = 3{,}840$ attacked scenarios. The present multi-domain design deliberately broadens the attack taxonomy and opinion state space relative to an initial single-domain study (100 profiles $\times$ 4 attacks $\times$ 4 Defence-only opinion leaves) to test whether personality–susceptibility signals observed in a single-domain, conventional-attack design generalise across domains and to AI-generated adversarial content.

Threat Model

The threat model corresponds to a state-level or well-resourced non-state adversary with the following capabilities and objectives:

Adversary capabilities. The adversary can generate large volumes of targeted persuasive content across multiple platforms simultaneously, personalise attack content to individual psychological profiles (obtained through data brokerage, social media scraping, or inferential modelling), sustain coordinated amplification campaigns to create false social consensus, and now deploy AI-generated synthetic media (deepfakes, AI images, personalised chatbot interactions) that lower the detection threshold for manipulated content.

Adversary objectives. The adversary seeks to shift a target population's political opinions on specific issues—alliance commitments, defence spending, platform regulation, immigration, geopolitical alignment—in a direction that serves strategic goals, regardless of whether the shifted opinion is accurate or beneficial for the target population.

Scope of this study. We model adversarial effectiveness at the *individual opinion level*: for a given profile, attack mechanism, and opinion dimension, does exposure produce signed opinion movement in the adversary's intended direction? We do not model: coordinated campaign dynamics, within-person belief-system coherence constraints, social network effects, or temporal stability of induced opinion shifts.

Defender use case. Security analysts and policy institutions can use the pipeline

to audit population segments: which psychographic profiles are most susceptible to which attack types on which issues? This supports targeted protective interventions—inoculation content calibrated to the most vulnerable profiles and the highest-risk issue areas.

Results

Descriptive Overview

The present study generated $80 \times 6 \times 8 = 3{,}840$ attacked scenarios across 80 pseudo-profiles, six attack vectors (Misleading Narrative Framing, Fear Appeal Scapegoating Post, Astroturf Comment Wave, Pseudo Expert Authority Cue, LLM Chatbot Personalised Persuasion, Deepfake Audio Speech Impersonation), and eight opinion leaves sampled across all four top-level opinion domains: Defence and National Security (Alliance Commitment Support, Defense Industrial Reshoring Support), Foreign Policy and Geopolitics (Diplomacy First Preference, Overseas Development Aid Support), Immigration and Citizenship (Birthright Citizenship Support, Integration Funding Support), and Information Integrity and Platforms (Counter Disinformation Agency Support, Trust in Mainstream Journalism). Full configuration parameters are reported in Table 1.

Mean absolute opinion displacement was $|\bar{\Delta}| = 34.75$ (SD $= 20.19$) on the $[-1000, +1000]$ scale. Mean signed shift was $\overline{\Delta} = +0.62$ and mean adversarial effectivity was $\overline{\mathrm{AE}} = -0.84$ (SD $= 40.18$), with 48.2% of scenarios producing positive AE (Table 2). Both the near-zero signed shift and the near-chance AE positive-rate indicate that, averaged over the broader multi-domain attack surface, exposure produced no systematic movement in the adversarial direction: substantial opinion change occurred, but in directions that were effectively random with respect to the adversarial goal.

The absence of a uniform boomerang pattern is a genuine departure from the single-domain predecessor study (Defence-only), in which three of four opinion leaves showed systematic resistance-direction shifts. Per-leaf AE ranges were tight in the present design ($\overline{\mathrm{AE}}_{\mathrm{leaf}} \in [-2.79, +1.30]$, all SDs $\in [39, 42]$), and absolute displacements were similar across all eight leaves ($|\bar{\Delta}| \in [33.87, 36.18]$; Table 2)—consistent with a ceiling on leaf-level heterogeneity once the opinion state space is sampled across unrelated policy domains.

Variance Partitioning

Intraclass correlation (Shrout and Fleiss, 1979) at the profile level was ICC(1) $= 0.000$ for absolute displacement ($\sigma^2_{\mathrm{between}} = 0.00$, $\sigma^2_{\mathrm{within}} = 409.06$, $k = 80$ profiles, cluster size $= 48.0$), 0.000 for signed displacement ($\sigma^2_{\mathrm{within}} = 1{,}618.03$), and 0.000 for adversarial effectivity ($\sigma^2_{\mathrm{within}} = 1{,}616.00$). Point-estimated between-profile variance collapses to zero across all three outcomes in the present analysis.

This variance structure sets a hard upper bound on profile-level moderation in the broader design: essentially none of the scenario-level variance in opinion shift magnitude or direction partitions to stable between-profile differences once a single profile is observed across 48 unrelated attack–opinion task cells. The result is consistent with persuasion research showing that context-level factors dominate individual differences (Matz et al., 2017), and it motivates treating the attack–opinion task rather than the profile as the

primary unit of moderation analysis.

Attack–Opinion Task Heterogeneity

Across the 48 attack–opinion task cells, ridge regression was fit separately for each cell (5-fold cross-validated α grid; five α candidates in $[10^1, 10^{3.5}]$). Cross-validated MSE ranged from 1,189.6 (Fear Appeal × Alliance Commitment Support) to 1,969.9 (Astroturf × Overseas Development Aid Support); reliability weights $w_t = n_t/\text{CV-MSE}_t$ after normalisation spanned $\tilde{w} \in [0.017, 0.029]$, a much narrower range than the analogous 16-cell design in the single-domain predecessor study. The narrower weight range is diagnostic: once a profile is observed across both conventional and AI-generated attacks on both Defence and non-Defence domains, no single task systematically distinguishes profiles from one another.

Structural Equation Model

The repeated-outcome SEM simultaneously regressed attack-averaged AE for each of the eight opinion leaves on six profile predictors (Neuroticism, Openness, Conscientiousness, Sex Female, Sex Other, mean baseline absolute score), with freely estimated residual covariances across all $\binom{8}{2} = 28$ indicator pairs. Estimation used `semopy` with the default MLW estimator; the model converged with near-saturated fit: $\chi^2(21) = 0.025$, $p = 1.00$; CFI = 1.000; GFI = 0.9997; AGFI = 0.9988; TLI = 1.000; RMSEA = 0.000; AIC = 168.0; BIC = 368.1. As in the single-domain predecessor study, the near-saturated fit indicates that at $N = 80$ the model is effectively unfalsifiable, and fit statistics should not be interpreted as confirmation of model quality.

No profile predictor reached $p < .05$ for any of the 48 leaf-specific paths; four paths reached $p < .10$ (all uncorrected): Integration Funding Support ← Conscientiousness ($\hat{\beta} = -3.30$, $p = .061$), Trust in Mainstream Journalism ← Conscientiousness ($\hat{\beta} = -2.99$, $p = .073$), Defense Industrial Reshoring Support ← Neuroticism ($\hat{\beta} = -3.13$, $p = .095$), and Defense Industrial Reshoring Support ← mean baseline absolute score ($\hat{\beta} = -3.21$, $p = .092$). The Conscientiousness paths share the negative sign observed in the single-domain predecessor study, but none survives FDR correction across the 48-path family. Full SEM path coefficients are in Supplementary Table S5.

Profile Moderator Analysis with FDR Correction

The profile-level OLS model (outcome: mean AE per profile, $n = 80$) with HC3 heteroscedasticity-robust errors explained $R^2 = 0.151$ (Adj. $R^2 = 0.055$; $F(8, 71) = 1.612$, $p = .137$). The model was not jointly significant.

No predictor survived Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995) across the moderator family (Table 3). Extraversion was the only predictor whose block-bootstrap 95% CI excluded zero ($\hat{\beta} = +1.61$, SE = 0.71, $p = .023$; bootstrap 95% CI $[0.47, 3.00]$; $B = 600$). Conscientiousness retained the negative sign observed in the single-domain predecessor study but attenuated and fell below significance ($\hat{\beta} = -1.06$, $p = .159$; bootstrap 95% CI $[-2.46, 0.30]$); Neuroticism ($\hat{\beta} = -0.86$, $p = .220$), Openness ($\hat{\beta} = +0.13$, $p = .862$), Agreeableness ($\hat{\beta} = -0.01$, $p = .989$), both sex dummies, and mean baseline absolute score were all non-significant at any conventional level.

The positive Extraversion coefficient indicates that more outgoing, socially expressive

profiles showed nominally higher adversarial effectivity scores in the multi-domain design—a susceptibility-amplifying direction. Because the CI excludes zero only marginally at the unadjusted threshold, because this predictor was not pre-registered as a primary hypothesis, and because the moderator-family FDR adjustment returns $q > .05$, we treat Extraversion as an exploratory finding in need of replication, not a confirmed moderator.

Hierarchical Importance Decomposition

Leave-one-group-out cross-validated R^2 decomposition computed marginal contributions of 20 ontology-aligned feature groups to the conditional susceptibility index. Full model CV-$R^2 \approx -0.028$, confirming that out-of-sample predictive power is essentially nil at $N = 80$: overfitting cost exceeds in-sample explained variance by a small margin.

Within-sample relative importance (absolute marginal R^2, normalised) attributed the largest shares to the Big Five collectively (22.1%), the Political Engagement Inventory (10.1%), the Need for Cognition dimension of the Dual Process Inventory (8.9%), the Platform Mechanism Awareness dimension of the Digital Literacy Inventory (7.3%), and smaller shares to individual Big Five facets (Openness 5.8%; Extraversion 5.8%; Conscientiousness 0.1%). Demographics contributed 2.7% of normalised relative importance. The near-zero CV-R^2 emphasises that the susceptibility index should be interpreted as a conditional relative ranking within the present design and its configured target set, not an absolute predictive instrument; validation against held-out or human data is required (see Construct Validity).

Bootstrap Confidence Intervals on Susceptibility Rankings

Profile-level block bootstrap resampling ($B = 600$) yielded 95% confidence intervals on all primary OLS coefficients (Table 3; full parameters in `bootstrap_primary_params.csv`). Only the Extraversion interval excluded zero ($[0.47, 3.00]$, bootstrap mean 1.66, bootstrap SD 0.66); all other coefficients had bootstrap CIs overlapping zero. At the profile rank level, percentile-ranked susceptibility indices clustered narrowly around the median, with the top five most-susceptible profiles showing mean AE between +0.08 and +10.67—an order of magnitude smaller than the leaf-level standard deviations, confirming that profile-level susceptibility rankings are descriptive rather than predictive at the current study scale.

Discussion

Attack–Opinion Context Dominates Profile-Level Variance

The collapse of ICC(1) to effectively zero across all three outcomes (absolute displacement, signed displacement, adversarial effectivity) in the present analysis confirms that the combination of attack vector and opinion leaf is the primary determinant of outcome variance once the opinion state space is sampled across multiple policy domains: no detectable share of scenario-level variance partitions to stable between-profile differences. This is a stronger version of the result from the single-domain predecessor study (ICC(1) = 0.052 for absolute displacement), and it aligns with the persuasion literature showing that psy-

Figure 1: Factorial study design. Six attack vectors (Misleading Narrative Framing, Fear Appeal Scapegoating Post, Astroturf Comment Wave, Pseudo Expert Authority Cue, LLM Chatbot Personalised Persuasion, Deepfake Audio Speech Impersonation) crossed with eight opinion leaves sampled across four top-level domains (Defence and National Security; Foreign Policy and Geopolitics; Immigration and Citizenship; Information Integrity and Platforms), for $80 \times 6 \times 8 = 3{,}840$ attacked scenarios.

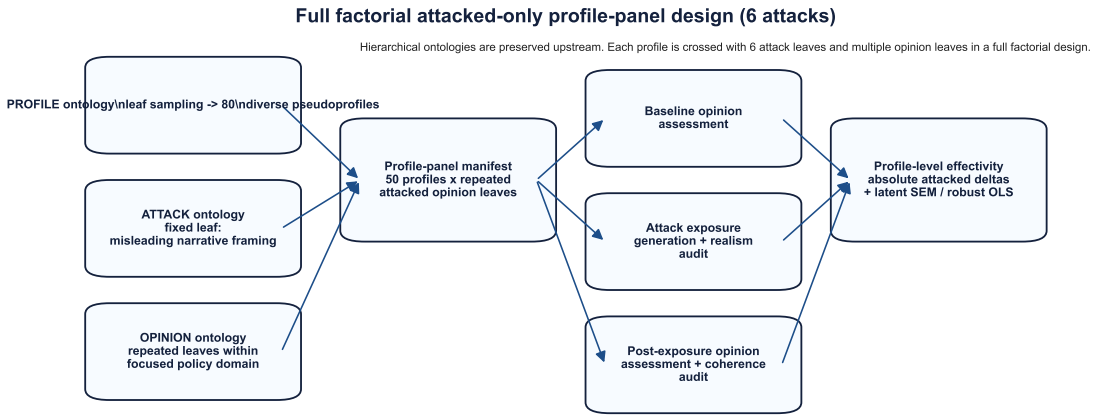


Table 1: Study design and configuration for the full-factorial profile-panel simulation.

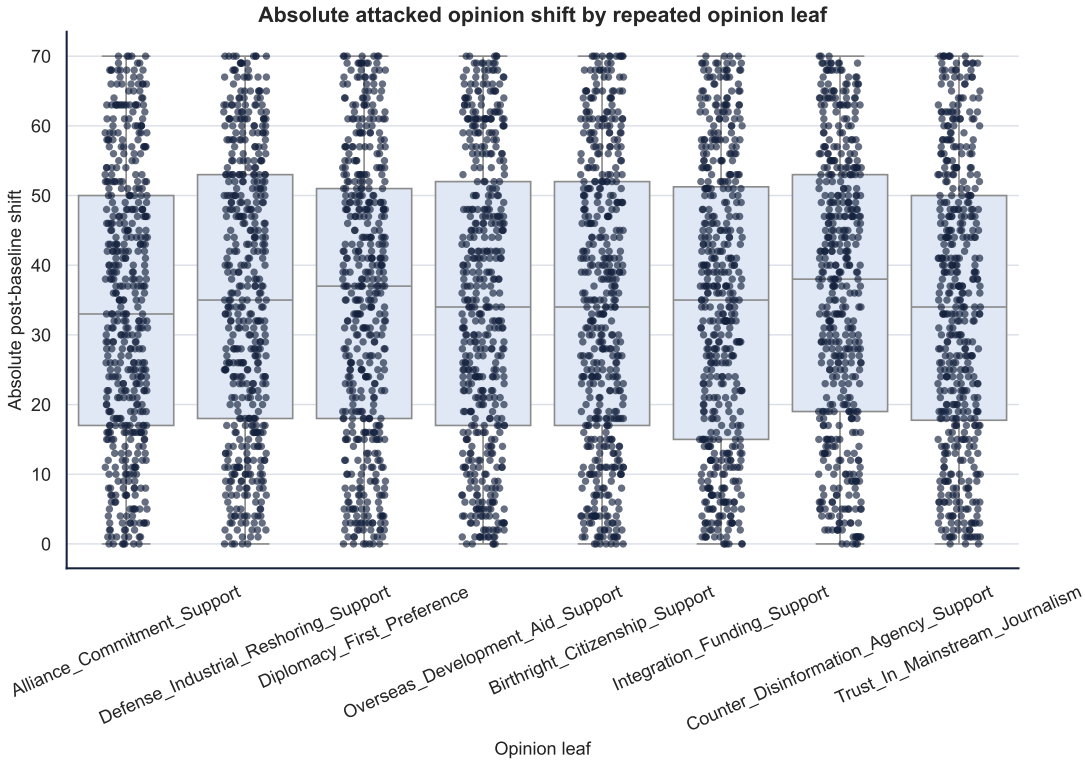
Parameter	Value
Run ID	run_9
Paper title	Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimensional State Space of Political Opinions
Profiles	80
Attacked rows	3840
Attack leaves	6 (Misleading_Narrative_Framing, Fear_Appeal_Scapegoating_Post, Astroturf_Comment_Wave, Pseudo_Expert_Authority_Cue, LLM_Chatbot_Personalized_Persuasion, Deepfake_Audio_Speech_Impersonation)
Opinion domain	Multiple
Repeated opinion leaves	8
Design	Full factorial (80 x 6 x 8)
Model	mistralai/mistral-small-3.2-24b-instruct

Note. Full factorial design: each profile is crossed with 6 attack leaves and 8 opinion leaves. All rows are attacked; there is no no-attack control condition.

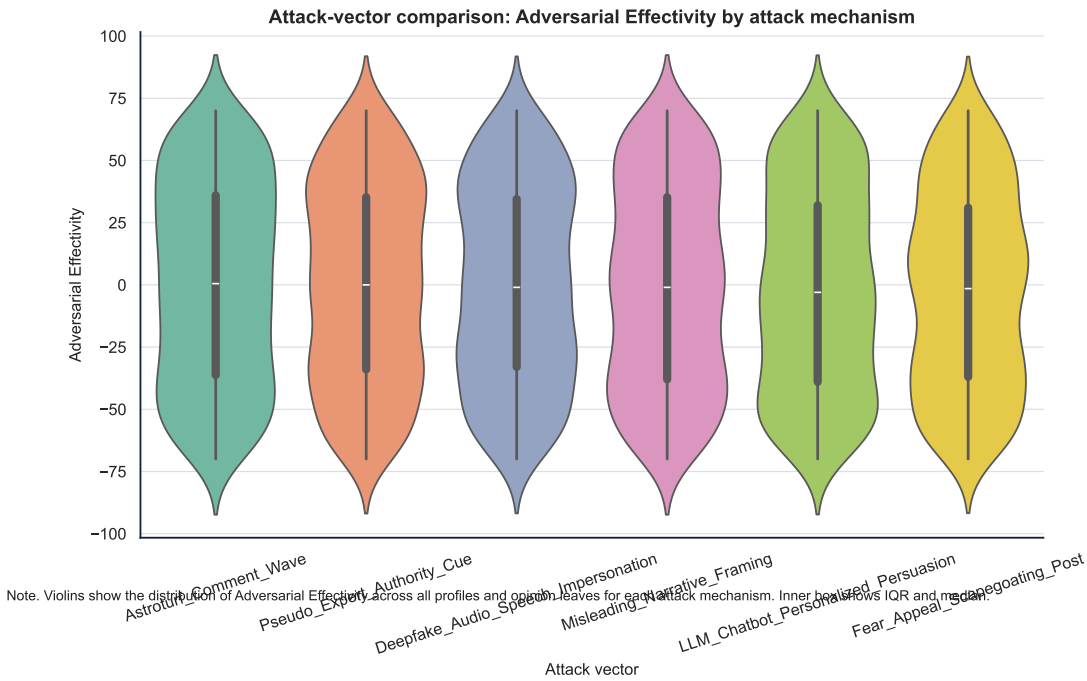
Table 2: Attacked effectivity descriptive statistics by repeated opinion leaf.

opinion_leaf_label	n_rows	mean_baseline	mean_post	mean_signed_delta	mean_abs_delta	sd_abs_delta	mean_adv_eff	sd_adv_eff
Counter_Disinformation_Agency_Support	480	347.092	348.929	1.837	36.179	19.947	-1.837	41.273
Defense_Industrial_Reshoring_Support	480	340.835	340.621	-0.215	35.506	20.048	0.215	40.775
Diplomacy_First_Preference	480	309.962	309.048	-0.915	35.048	20.249	-0.915	40.466
Overseas_Development_Aid_Support	480	201.906	203.129	1.223	34.777	20.631	-1.223	40.418
Birthright_Citizenship_Support	480	169.071	171.535	2.465	34.281	20.577	-2.465	39.907
Integration_Funding_Support	480	221.494	220.194	-1.300	34.267	20.367	1.300	39.841
Alliance_Commitment_Support	480	283.167	285.956	2.790	34.069	19.553	-2.790	39.182
Trust_In_Mainstream_Journalism	480	217.690	216.721	-0.969	33.873	20.009	0.969	39.329

Note. Descriptive statistics aggregated across 6 attack vectors per opinion leaf. Adversarial effectivity weights shift by the per-leaf adversarial goal direction.



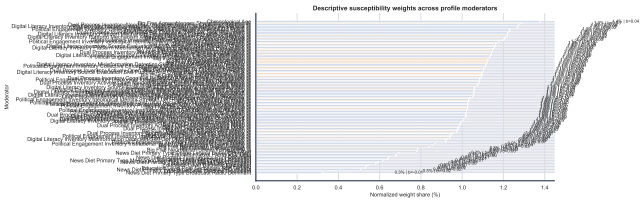
(a) Panel A: Opinion displacement distribution. Each point represents one scenario (n = 3,840). Colour encodes attack vector. Opinion displacement was large and similar across all six attack types and all eight opinion leaves, while adversarial effectivity (directional) was near zero and similarly distributed across leaves.



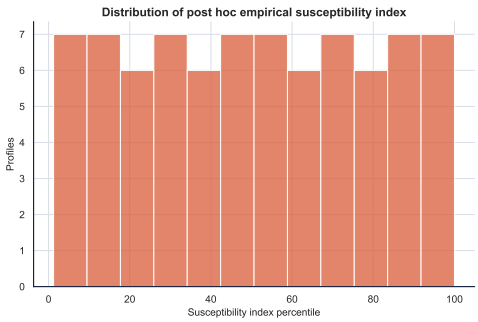
Note. Violins show the distribution of Adversarial Effectivity across all profiles and opinion leaves for each attack mechanism. Inner boxplots shows IQR and median.

(b) Panel B: Adversarial effectivity by attack vector. Violin plots show the full distribution across all profiles and opinion leaves for each attack mechanism; no attack type produces systematically positive mean adversarial effectivity.

Figure 2: Opinion displacement and adversarial effectivity. Panel A shows absolute opinion shifts across scenarios; Panel B compares adversarial effectivity distributions by attack vector. Together they confirm that substantial opinion change occurred in directions effectively random with respect to the adversarial goal, and that no single attack mechanism shows systematic dominance.



(a) Panel A: Profile-moderator coefficient forest. Points show OLS coefficient estimates on profile-mean adversarial effectivity with HC3 robust errors; horizontal bars show block-bootstrap 95% CIs ($B = 600$). Only the Extraversion interval excludes zero at the uncorrected level; no coefficient survives Benjamini–Hochberg FDR correction.



(b) Panel B: Susceptibility index distribution. Histogram of S_i^* across all profiles with 90% bootstrap confidence intervals shown for the top and bottom quintile profiles.

Figure 3: Profile-level moderation and susceptibility. Panel A shows moderator coefficient estimates with bootstrap uncertainty; Panel B shows the distribution of conditional susceptibility indices across profiles. Both panels confirm that profile-level variance is small and poorly discriminated at the present sample size.

Table 3: Profile moderators of attacked effectivity: controlled contrasts, SEM paths, and descriptive susceptibility weights.

Moderator	Role	Controlled mean b	Controlled p	Ridge mean b	Weight %	SEM mean b	SEM mean |b|	SEM min p	SEM paths
Big Five Conscientiousness Mean %	core	-1.034	0.123			-1.185	2.010	0.061	8.000
Big Five Neuroticism Mean %	core	-0.934	0.134			-1.039	1.560	0.095	8.000
Sex Other	core	-0.429	0.780	0.002	0.779	-0.776	1.952	0.237	8.000
Big Five Openness To Experience Mean %	core	0.077	0.916			-0.016	1.391	0.114	8.000
Sex Female	core	0.485	0.741	-0.000	0.534	-0.259	2.512	0.162	8.000
Big Five Extraversion Mean %	exploratory	1.757	0.012						
Big Five Agreeableness Mean %	exploratory	-0.078	0.910						

Note. Controlled coefficients summarize moderator associations with mean attacked effectivity. SEM columns summarize the repeated-outcome path model across the attacked opinion leaves. Weight percentages come from the target-conditional regularized aggregation used to compute the post hoc susceptibility index.

chologically targeted content produces modest average individual-level effects (Bail et al., 2018; Matz et al., 2017). It has a direct operational implication: the same individual may resist one attack on one issue while being substantially moved by a different combination, and aggregate profile-based susceptibility classification therefore has limited utility as a *universal* vulnerability score. The appropriate framing is *conditional susceptibility* indexed to specific attack–opinion contexts.

Unlike the single-domain predecessor study, the present broader design did not exhibit a systematic boomerang pattern across domains: mean AE per leaf stayed within ± 3 units of zero, with no leaf producing a directionally large effect in either the adversarial or resistance direction. Rather than a substantive claim that adversarial attacks "work no better than chance", this pattern most plausibly reflects that the previously observed Defence-domain boomerang was an interaction between a particular adversarial-direction assignment and strongly anchored pro-defense positions in the underlying LLM—an interaction that does not generalise once non-defence domains are added to the target set.

Non-Replication of the Conscientiousness Signal

The most consequential scientific result of the present study is that the Conscientiousness effect observed in the single-domain predecessor study ($\hat{\beta} = -3.26$, $p = .004$, bootstrap 95% CI $[-5.10, -1.20]$, FDR $q = .030$) does *not* replicate in the extended design. In the present, broader multi-domain design, Conscientiousness retains its negative sign in both the OLS profile model ($\hat{\beta} = -1.06$, $p = .159$, bootstrap 95% CI $[-2.46, 0.30]$) and in several leaf-specific SEM paths (Integration Funding $p = .061$; Trust in Mainstream Journalism $p = .073$), but the effect size attenuates by roughly two-thirds and the interval crosses zero.

Two factors plausibly contribute. First, the present study halves the Defence-domain weight in the target set (from 4/4 leaves to 2/8) and adds AI-driven synthetic-media attacks that did not appear in the single-domain predecessor study; if the protective effect of deliberative processing is strongest against traditional narrative framing on defence policy—as hinted by the predecessor study's leaf-specific path coefficients—dilution across heterogeneous tasks will attenuate a profile-level average. Second, $N = 80$ profiles leaves the design underpowered for detecting a predecessor-study-magnitude effect after FDR correction across 8 moderators and 48 leaf-specific SEM paths; the $R^2 = 0.151$ and $F(8, 71) = 1.612$, $p = .137$ overall model indicates that the design provides weak discrimination between any of the profile predictors.

We therefore do not treat Conscientiousness as a disconfirmed moderator; we treat it as not replicated at FDR-corrected significance in a broader but less powered design. The appropriate inference is that *single-domain estimates of personality–susceptibility associations do not straightforwardly generalise to broader attack and opinion surfaces*, and that larger profile panels ($N \gtrsim 500$) are required to detect predecessor-magnitude moderators once the test family is FDR-corrected across a multi-domain, multi-attack design.

Extraversion as an Exploratory Susceptibility-Amplifying Finding

The one predictor whose bootstrap 95% CI excludes zero in the present analysis—Extraversion ($\hat{\beta} = +1.61$, $p = .023$, CI $[0.47, 3.00]$)—was not pre-registered as a primary moderator and was not significant in the single-domain predecessor study ($\hat{\beta} = -2.35$,

$p = .130$, with the opposite sign). The sign reversal across the two designs and the failure to survive FDR correction both argue against treating this as a confirmed finding. A plausible exploratory interpretation is that Extraversion, which captures social engagement, assertiveness, and sociability (McCrae and Costa, 1997), may increase receptivity to social-consensus and astroturfing-style attacks in contexts where the adversarial direction is aligned with socially shared positions. This interpretation is tentative and requires both replication and domain-conditional analysis.

Digital Literacy and AI-Based Attacks: No Detectable Domain-Specific Effects

The present study included two AI-driven attack vectors (Deepfake Audio Speech Impersonation, LLM Chatbot Personalised Persuasion) and a new Information Integrity opinion domain (Counter Disinformation Agency Support, Trust in Mainstream Journalism), and extended the profile ontology with Digital Literacy and Dual Process Inventories. The a-priori hypothesis, derived from experimental evidence on AI-personalised content (Goldstein et al., 2023; Guess et al., 2020; Kozyreva et al., 2020; Roozenbeek et al., 2022), was that Digital Literacy dimensions would show protective effects specifically for Information Integrity leaves targeted by synthetic media and AI-personalised attacks.

This hypothesis is not supported in the present analysis. The Digital Literacy Inventory accounts for only 2.2% of normalised relative importance in the hierarchical decomposition (with a Platform Mechanism Awareness sub-dimension at 7.3%), and neither the Trust in Mainstream Journalism nor the Counter Disinformation Agency Support leaf produces leaf-specific SEM paths reaching $p < .05$ for any Digital Literacy or Dual Process Inventory predictor. The most restrained interpretation is that the present sample size is insufficient to detect domain-conditional literacy effects once the test family is FDR-corrected, not that such effects are absent. A confirmatory test would require at least an order of magnitude more profiles and explicit pre-registration of the domain-conditional hypothesis.

Construct Validity

The simulation pipeline faces three construct validity constraints that must be stated explicitly.

First, *synthetic profiles are not humans.* LLM-generated pseudoprofiles produce conditioned responses that approximate human response distributions at aggregate levels (Argyle et al., 2023) but are not validated against individual-level criterion data. Personality–susceptibility associations in the simulation may diverge from human experimental findings in magnitude, direction, or both. Criterion validation against human participants exposed to analogous attack content is the necessary next step.

Second, *generated attack content is not real attack traffic.* The pipeline generates plausible platform-native manipulative content, audited for realism, but it is generated in a controlled context rather than collected from live adversarial operations. The attack content distribution may not match what adversaries actually deploy, and effectiveness estimates may be biased toward or against any specific attack type as a result.

Third, *adversarial-direction assignments embed researcher assumptions.* Each opinion leaf is assigned a fixed adversarial goal direction ($d_k \in \{-1, +1\}$), operationalising a specific assumption about adversarial intent. Changing direction assignments would change

adversarial effectivity scores. The assignments in the present study reflect a theoretically motivated model of what adversaries in the defence and geopolitical space are likely to pursue, but they should be made explicit and subjected to external review for any operational application.

Limitations

The cross-validated R^2 for the conditional susceptibility index is near zero (≈ -0.028) at the present scale ($N = 80$ profiles), confirming that aggregate profile-level prediction is effectively nil out-of-sample in the broader multi-domain design. In-sample $R^2 = 0.151$ is both small in absolute terms and not jointly significant against the null ($F(8, 71) = 1.612$, $p = .137$); the present study is best understood as a methodology and infrastructure demonstration rather than a definitive moderation study. Standard errors do not fully account for the clustered structure of repeated observations per profile; profile-level block bootstrap CIs ($B = 600$) are provided as a partial remedy (Cameron et al., 2008). The SEM fit indices ($\chi^2(21) = 0.025$, $p = 1.0$; CFI = 1.0; RMSEA = 0.0) approach perfect-fit territory, indicating the model is near-saturated at $N = 80$; fit statistics should not be interpreted as confirmation of model quality at this scale. The factorial design uses only attacked scenarios; a no-attack control condition would strengthen causal inference about exposure versus baseline opinion drift. The ICC(1) ≈ 0 estimate indicates that, with the current profile and task panel, no detectable between-profile variance in opinion shift survives the within-profile task-level noise; this is itself a substantive limitation on the feasibility of profile-level susceptibility discrimination in designs of comparable breadth and scale. The apparent non-replication of the Conscientiousness signal from the single-domain predecessor study is consistent with both a true attenuation of the effect across domains and a power-limited failure to detect an effect of similar magnitude; further studies with $N \gtrsim 500$ profiles and explicit domain stratification are required to distinguish these explanations.

Conclusion

The present study ($N = 3{,}840$ scenarios across 80 profiles, 6 attack vectors including two AI-driven synthetic-media attacks, 8 opinion leaves spanning four top-level domains) demonstrates that a full factorial, ontology-driven simulation produces an internally consistent, analytically decomposable measurement architecture for studying susceptibility to adversarial opinion manipulation. The central empirical result is negative: no profile-level predictor survives FDR correction across the moderator family, the between-profile variance component collapses to zero for all three outcomes, and the Conscientiousness signal observed in the single-domain predecessor study (Defence-only) does not replicate at FDR-corrected significance in the broader multi-domain, AI-inclusive design. Extraversion emerges as the only predictor whose bootstrap 95% CI excludes zero ($\hat{\beta} = +1.61$, CI $[0.47, 3.00]$), but with the opposite sign to the predecessor study and without pre-registration or FDR-corrected significance, and it is treated as exploratory.

This pattern is methodologically informative. It suggests that single-domain

personality–susceptibility claims are fragile to design breadth: the apparent dominance of Conscientiousness in the single-domain predecessor study was either a domain-specific effect that attenuates once non-defence issue domains are included, or an effect of sufficient magnitude that a power increase to $N \gtrsim 500$ profiles is required to detect it under FDR correction in a multi-domain family. Both interpretations have direct implications for operational susceptibility auditing: any deployed profile-based vulnerability classifier must be validated on the specific attack–opinion surface it is claimed to characterise, not extrapolated from narrower single-domain studies.

The five principal contributions of the pipeline infrastructure are: (1) adversarial effectivity as a directional primary outcome, distinguishing successful manipulation from resistance and boomerang effects; (2) conditional susceptibility indexing with cross-task ridge aggregation and reliability-weighted task contributions; (3) profile-level block-bootstrap confidence intervals quantifying uncertainty in profile coefficients and susceptibility rankings; (4) a fully generic CLI-accessible pipeline supporting user-defined ontologies for deployment in custom threat contexts, including an extended profile ontology spanning Big Five, Dual Process, Digital Literacy, and Political Engagement inventories; (5) an extended attack taxonomy covering AI-generated synthetic media and personalised LLM persuasion alongside four conventional social-media attack mechanisms.

Substantive claims about real populations require validation against human experimental data; the present study provides the methodological and software infrastructure for that comparison. The pipeline is publicly available and designed for modular extension to additional attack categories, psychological inventories, and opinion domains as the threat landscape evolves.

Dual-Use Ethics Statement

This research models mechanisms of cognitive vulnerability to adversarial persuasion. The primary intended application is **defensive**: identifying population segments for which particular attack–opinion combinations produce elevated susceptibility, enabling targeted inoculation, media literacy interventions, and institutional protection strategies.

We acknowledge the dual-use risk: the same susceptibility estimates that enable defence also characterise exploitable targets. We mitigate this risk through the following design decisions: (i) all profiles are synthetic and not linked to identifiable individuals; (ii) the output of the pipeline is population-level susceptibility patterns indexed to psychographic profile dimensions, not a tool for identifying and targeting specific individuals; (iii) the pipeline outputs conditional estimates that are weak at the individual level (near-zero aggregate R^2), limiting precision-targeting utility; (iv) the code and methodology are published openly to enable independent validation, critique, and controlled deployment.

Any deployment in an operational context should be subject to institutional ethics review, restricted to defence-oriented applications, and should explicitly prohibit use for adversarial population targeting.

Reproducibility Statement

The full simulation pipeline is implemented in Python and available at the project repository. The pipeline is fully reproducible from the ontology JSON files and the random seed; all LLM calls are logged with model slug, temperature, and full prompt/response pairs. The pipeline configuration for the present study is recorded in `evaluation/run_9/config/pipeline_config.json`; the single-domain predecessor's configuration is available in `evaluation/run_8/config/`. The present study can be reproduced end-to-end by executing `scripts/run_9.sh`; the script drives the nine-stage pipeline (scenario generation, baseline elicitation, attack generation, post-exposure elicitation, effectivity deltas, SEM construction, research visualisations, publication assets, report build) against the shared ontology files in `src/backend/ontology/separate/test/`. All statistical analyses are implemented without external R dependencies; the SEM uses `semopy` (Python implementation compatible with Rosseel (2012)). Bootstrap resampling uses a fixed seed for deterministic reproduction (`seed=99` for the present study; `seed=42` for the single-domain predecessor's configuration). The stage-level artefacts underlying every number reported in this paper—delta summaries, ICC JSONs, OLS coefficient tables, SEM fit indices, bootstrap CSVs, ridge reliability weights, and hierarchical importance breakdowns—are persisted under `evaluation/run_9/stage_outputs/` with a per-stage `manifest.json` for hash-level verification.

Methods

Ontology-Constrained Factorial Design

The present study and the single-domain predecessor study use the same three hierarchical JSON ontologies; the two designs differ only in which leaves enter the factorial design.

PROFILE ontology. Continuous leaves are expressed as percentile scores (0–100). The present study uses 85 feature dimensions: Big Five personality (NEO-PI-R structure; five traits × 4–6 facets), chronological age, sex (categorical with three levels), and three additional psychometric inventories—the Dual Process Inventory (analytical thinking, cognitive effort, intuitive override, need for cognition), the Digital Literacy Inventory (critical evaluation, source triangulation, algorithmic awareness, platform mechanism awareness, misinformation detection), and the Political Engagement Inventory (news engagement, political efficacy, collective efficacy beliefs, civic participation, institutional trust, ideological openness). The single-domain predecessor study used the 42-dimension Big Five + demographics subset of the same ontology.

ATTACK ontology (18 leaves total). Leaves are organised under four mechanism families: Social Media Misinformation (Misleading Narrative Framing; Fabricated Statistic Claim; Selective Historical Analogy; Strawman Quote Fabrication); Coordinated Amplification (Astroturf Comment Wave; Bot-Like Repetition Burst; Review Bombing Cascade; Cross-Platform Narrative Laundering); Manipulative Persuasion (Fear Appeal Scapegoating Post; Ingroup–Outgroup Identity Priming; Pseudo Expert Authority Cue; False Flag Accusation; Urgency–Scarcity Priming); Synthetic Media Generation (Deepfake Audio Speech Impersonation; AI-Generated Image Event Fabrication; LLM Chatbot Personalized Persuasion). The present study selects six leaves: Misleading Narrative Framing, Fear Appeal Scapegoating Post, Astroturf Comment Wave, Pseudo Expert Authority Cue, LLM Chatbot Personalized Persuasion, Deepfake Audio Speech Impersonation. The single-domain predecessor study selected the first four.

OPINION ontology (62 leaves total). Leaves are organised under four top-level domains: Immigration and Citizenship (17 leaves); Foreign Policy and Geopolitics (17 leaves); Defence and National Security (16 leaves); Information Integrity and Platforms (12 leaves). Each leaf carries a pre-specified adversarial direction $d_k \in \{-1, +1\}$ encoding the operator's intended opinion movement. Of 62 leaves, 57 carry non-zero adversarial direction. The present study samples 8 leaves spanning all four domains (two per domain: Alliance Commitment Support and Defense Industrial Reshoring Support from Defence; Diplomacy First Preference and Overseas Development Aid Support from Foreign Policy; Birthright Citizenship Support and Integration Funding Support from Immigration; Counter Disinformation Agency Support and Trust in Mainstream Journalism from Information Integrity). The single-domain predecessor study sampled 4 leaves from Defence and National Security only.

Profile Generation

Profiles are generated deterministically by stratified sampling across the joint continuous-attribute space using percentile grids (profile_candidate_multiplier = 5). Categorical attributes (sex, education level, news diet) are assigned by nearest-neighbour matching in the continuous space. No LLM calls are used for profile generation; this ensures exact reproducibility and eliminates generation-side confounding.

Baseline and Post-Exposure Opinion Elicitation

For each profile i, attack a, and opinion leaf k: (1) Baseline opinion is elicited via a structured role-playing prompt encoding the profile's full attribute vector, the opinion leaf's domain context, and a request for a self-reported opinion score on the $[-1000, +1000]$ scale. (2) Attack content is generated as a platform-native manipulative message tailored to the specific attack–opinion–profile combination, with direction-aware persuasion goals derived from d_k and the profile's personality moderator values. (3) Post-exposure opinion is elicited with the same scoring prompt after presenting the attack content.

All LLM calls use Mistral Small (mistralai/mistral-small-3.2-24b-instruct) via the OpenRouter API at temperature = 0.15 with up to two realism/coherence repair iterations.

Adversarial Effectivity and Primary Outcome

Let $d_k \in \{-1, +1\}$ denote the pre-specified adversarial direction for leaf k. The signed shift for profile i under attack a on leaf k is $\Delta_{iak} = \text{post}_{iak} - \text{baseline}_{iak}$, and adversarial effectivity is $\text{AE}_{iak} = \Delta_{iak} \times d_k$. Positive AE means the attack moved the opinion in the adversarially intended direction; negative values indicate resistance or backfire.

Conditional Susceptibility Index

For each attack–opinion task $t = (a, k)$, a ridge regression model is fit on the n_t profile observations:

$$\hat{\beta}_t = \arg\min_{\beta} \sum_i (\text{AE}_{iak} - x_i^\top \beta)^2 + \alpha_t \|\beta_{-0}\|_2^2, \tag{1}$$

where α_t is selected by 5-fold cross-validation. The reliability weight for task t is $w_t = n_t/\text{CV-MSE}_t$. The conditional susceptibility index for profile i over target set $\mathcal{T}$ is the percentile rank of the reliability-weighted predicted adversarial effectivity:

$$S_i^*(\mathcal{T}) = \text{PctRank}\left(\sum_{t \in \mathcal{T}} \tilde{w}_t \, x_i^\top \hat{\beta}_t\right), \quad \tilde{w}_t = w_t \Big/ \sum_{t'} w_{t'}. \tag{2}$$

Optional cross-task partial pooling (Empirical-Bayes shrinkage, $\lambda \in [0, 1]$) shrinks each task's coefficient vector toward the reliability-weighted cross-task mean, reducing overfitting when a task has few observations. Block bootstrap ($B = 600$ resamples at profile level) provides 90% confidence intervals on percentile ranks and cross-task coefficient standard deviations.

Hierarchical Importance Decomposition

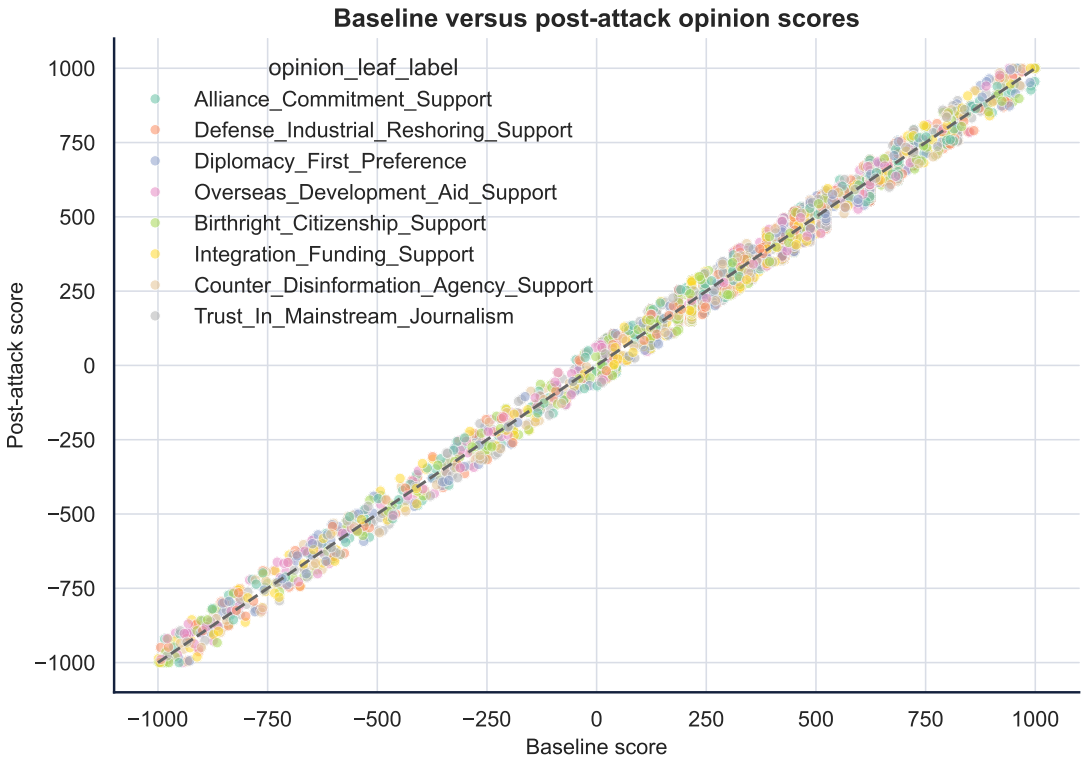
Feature groups are discovered from column names via prefix trie (no hard-coded inventory names). For each group g, marginal cross-validated R^2 is $\Delta R^2_g = R^2_{\text{full}} - R^2_{\text{ablated } g}$, where the ablated model omits group g's columns. Relative importance is $|\Delta R^2_g| / \sum_{g'} |\Delta R^2_{g'}| \times 100$.

Structural Equation Model

A profile-level repeated-outcome SEM (MacKinnon, 2008; Rosseel, 2012) simultaneously regresses attack-averaged adversarial effectivity for each opinion leaf on profile moderators, with freely estimated residual covariances. Estimation uses OLS with HC3 heteroscedasticity-robust covariance (profile-level bootstrap standard errors for key paths). FDR correction uses the Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995) applied to all moderator p-values.

Supplementary Materials

Figure 4: Supplementary Figure S1: Baseline versus post-attack opinion scores. Each point represents one (profile, attack, opinion-leaf) scenario. Colour encodes adversarial effectivity (positive = manipulation succeeded).



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References

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., and Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351. doi:10.1017/pan.2023.2.

Bail, C. A., Argyle, L. P., Brown, T. W., Bumpus, J. P., Chen, H., Hunzaker, M. B. F., Lee, J., Mann, M., Merhout, F., and Volfovsky, A. (2018). Exposure to opposing views

Figure 5: Supplementary Figure S2: Per-profile attack-effectivity heatmap. Rows are profiles ordered by susceptibility index (ascending); columns are attack–opinion task combinations. Colour encodes adversarial effectivity.

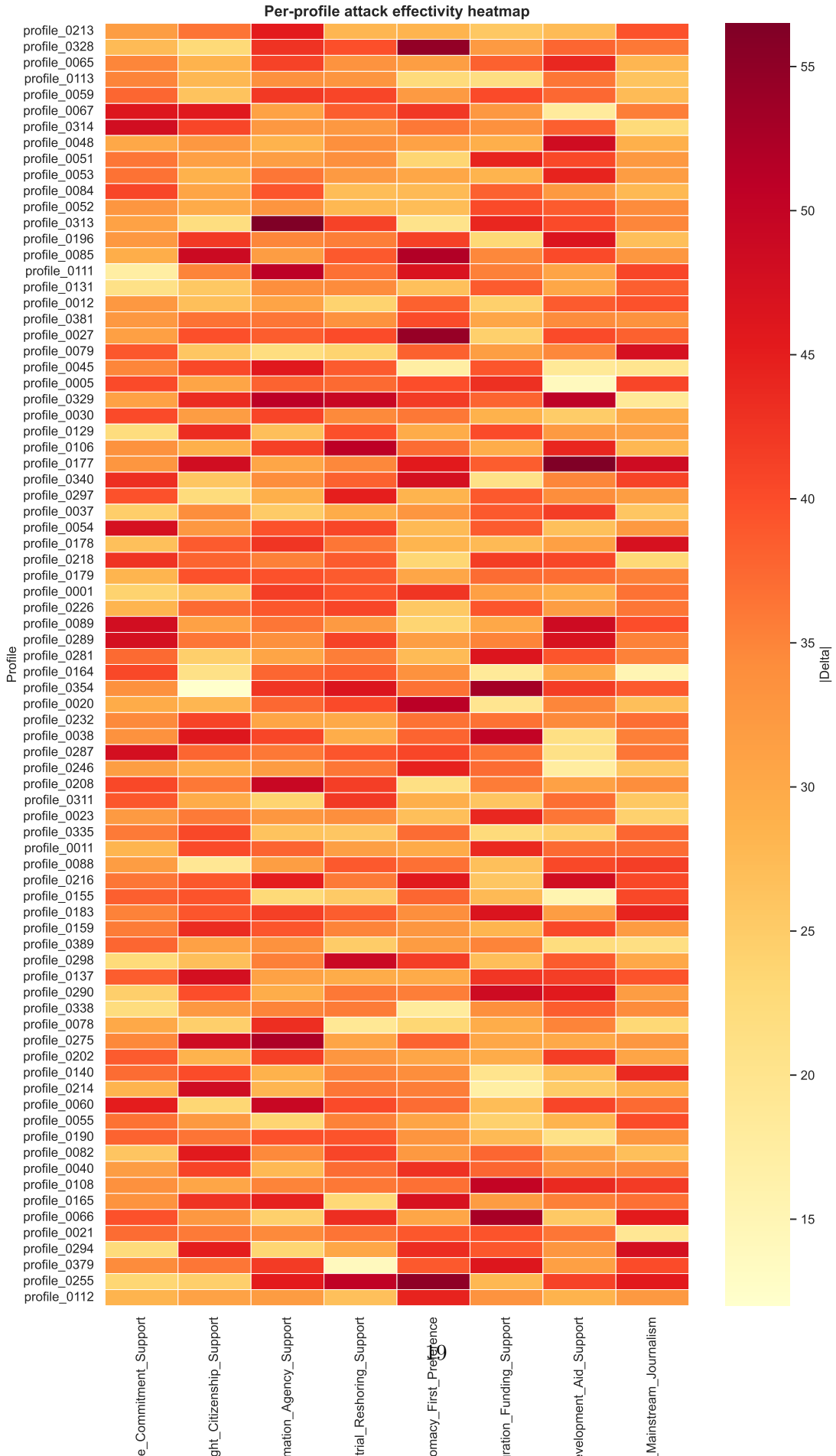

Table 4: Ontology leaves used in run_9.

Ontology	Role	Leaf
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Manipulative_Persuasion > Fear_Appeal_Scapegoating_Post
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Coordinated_Amplification > Astroturf_Comment_Wave
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Manipulative_Persuasion > Pseudo_Expert_Authority_Cue
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Synthetic_Media_Generation > LLM_Chatbot_Personalized_Persuasion
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Synthetic_Media_Generation > Deepfake_Audio_Speech_Impersonation
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Alliance_Commitment_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Defense_Industrial_Reshoring_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Foreign_Policy_and_Geopolitics > Diplomacy_First_Preference
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Foreign_Policy_and_Geopolitics > Overseas_Development_Aid_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Immigration_and_Citizenship > Birthright_Citizenship_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Immigration_and_Citizenship > Integration_Funding_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Information_Integrity_and_Platforms > Counter_Disinformation_Agency_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Information_Integrity_and_Platforms > Trust_In_Mainstream_Journalism

Note. Only ontology leaves are sampled for estimation. The ATTACK ontology contributes 6 factorial leaves; the OPINION ontology contributes repeated indicator leaves.

Table 5: Core and exploratory profile moderator comparison.

Moderator	Role	Multivariate b	Multivariate p	Univariate b	Univariate p	Ontology group	Normalized weight %
Big Five Conscientiousness Mean %	core	-1.062	0.159	-1.034	0.123	Personality: Conscientiousness	0.992
Big Five Neuroticism Mean %	core	-0.862	0.220	-0.934	0.134	Personality: Neuroticism	0.784
Sex Other	core	-0.577	0.725	-0.429	0.780	Demographics: Sex	0.779
Big Five Openness To Experience Mean %	core	0.133	0.862	0.077	0.916	Personality: Openness	1.074
Sex Female	core	-0.231	0.880	0.485	0.741	Demographics: Sex	0.534
Big Five Extraversion Mean %	exploratory	1.612	0.023	1.757	0.012	Personality: Extraversion	1.032
Big Five Agreeableness Mean %	exploratory	-0.010	0.989	-0.078	0.910	Personality: Agreeableness	1.125

Note. Core terms are entered into the latent SEM or primary multivariate profile model. Normalized weight percentages summarize each moderator's share of the total fitted importance after accounting for the moderator's observed variability.

Table 6: Assumption register and peer-review risk register.

Type	Item	S
Assumption	Sample size is sufficient for stable moderation estimates.	c
Assumption	Attack-only design is intentional and aligned with the moderation question.	c
Assumption	Opinion measurement has enough numeric resolution.	r
Assumption	Attack text realism is adequate.	r
Assumption	SEM converges with acceptable fit.	c
Assumption	LLM outputs are reproducible and auditable.	c
Critique	Synthetic LLM agents may not represent human cognition or causal response behavior.	a
Critique	Adversarial content may be unrealistic or too generic.	a
Critique	Model dependence on a single LLM could bias outputs.	a
Critique	Insufficient sample size inflates uncertainty and may destabilize SEM.	a
Critique	Attack-only designs cannot estimate the incremental effect of exposure relative to a no-attack counterfactual.	a

Note. Assumptions and design risks are surfaced explicitly to support replication, adversarial critique, and validation against human experimental data.

on social media can increase political polarization. *Proceedings of the National Academy of Sciences*, 115(37):9216–9221. doi:10.1073/pnas.1804840115.

Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B (Methodological)*, 57(1):289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.

Bennett, W. L. and Livingston, S. (2018). The disinformation order: Disruptive communication and the decline of democratic institutions. *European Journal of Communication*, 33(2):122–139. doi:10.1177/0267323118760317.

Bradshaw, S. and Howard, P. N. (2019). The global disinformation order: 2019 global inventory of organised social media manipulation. Technical report, Oxford Internet Institute, University of Oxford, Oxford.

Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427. doi:10.1162/rest.90.3.414.

Goldstein, J. A., Sastry, G., Musser, M., DiResta, R., Gentzel, M., and Sedova, K. (2023). Generative language models and automated influence operations: Emerging threats and potential mitigations. *arXiv preprint arXiv:2301.04246*. doi:10.48550/arXiv.2301.04246.

Guess, A. M., Lerner, M., Lyons, B., Montgomery, J. M., Nyhan, B., Reifler, J., and Sircar, N. (2020). A digital media literacy intervention increases discernment between mainstream and false news in the united states and india. *Proceedings of the National Academy of Sciences*, 117(27):15536–15545. doi:10.1073/pnas.1920498117.

Hirsh, J. B., Kang, S. K., and Bodenhausen, G. V. (2012). Personalized persuasion: Tailoring persuasive appeals to recipients' Big Five personality traits. *Psychological Science*, 23(6):578–581. doi:10.1177/0956797611436349.

Table 7: Reproducibility manifest for run_9.

Field	Value
run_id	run_9
paper_title	Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimensional State Space of Political Opinions
output_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/evaluation
report_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/research_r
report_assets_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/research_r
n_scenarios	10
n_profiles	80
seed	99
attack_ratio	1.000
attack_leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
attack_leaves	Misleading_Narrative_Framing,Fear_Appeal_Scapegoating_Post,Astroturf_Comment_Wave,Psc
focus_opinion_domain	
max_opinion_leaves	8
profile_candidate_multiplier	5
primary_moderator	posthoc_profile_susceptibility_index
bootstrap_samples	600
use_test_ontology	True
ontology_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/src/backen
openrouter_model	mistralai/mistral-small-3.2-24b-instruct
temperature	0.150
max_repair_iter	2
profile_generation_mode	deterministic
self_supervise_attack_realism	True
realism_threshold	0.760
self_supervise_opinion_coherence	True
coherence_threshold	0.760
generate_visuals	True
export_static_figures	True
build_report	True
resume_from_stage	06
stop_after_stage	09
save_raw_llm	True
timeout_sec	90
max_concurrency	20
run_stage_checks	False
stage_check_scenarios	3

Note. The manifest captures the full pipeline configuration used to generate all simulation outputs and manuscript assets reported in this paper.

Table 8: Leaf-specific path-SEM coefficients.

Outcome leaf	Moderator	SEM b	
Adversarial Delta Indicator Alliance Commitment Support	Big Five Neuroticism Mean %	-1.430	1.
Adversarial Delta Indicator Alliance Commitment Support	Big Five Openness To Experience Mean %	-2.414	1.
Adversarial Delta Indicator Alliance Commitment Support	Big Five Conscientiousness Mean %	-1.335	1.
Adversarial Delta Indicator Alliance Commitment Support	Sex Female	-0.271	4.
Adversarial Delta Indicator Alliance Commitment Support	Sex Other	1.808	4.
Adversarial Delta Indicator Alliance Commitment Support	Mean Baseline Abs Score	-2.609	1.
Adversarial Delta Indicator Birthright Citizenship Support	Big Five Neuroticism Mean %	-1.291	1.
Adversarial Delta Indicator Birthright Citizenship Support	Big Five Openness To Experience Mean %	0.035	1.
Adversarial Delta Indicator Birthright Citizenship Support	Big Five Conscientiousness Mean %	0.398	1.
Adversarial Delta Indicator Birthright Citizenship Support	Sex Female	-1.836	4.
Adversarial Delta Indicator Birthright Citizenship Support	Sex Other	-0.177	4.
Adversarial Delta Indicator Birthright Citizenship Support	Mean Baseline Abs Score	-0.697	1.
Adversarial Delta Indicator Counter Disinformation Agency Support	Big Five Neuroticism Mean %	-0.281	1.
Adversarial Delta Indicator Counter Disinformation Agency Support	Big Five Openness To Experience Mean %	1.782	1.
Adversarial Delta Indicator Counter Disinformation Agency Support	Big Five Conscientiousness Mean %	-2.404	1.
Adversarial Delta Indicator Counter Disinformation Agency Support	Sex Female	-5.839	4.
Adversarial Delta Indicator Counter Disinformation Agency Support	Sex Other	-2.357	3.
Adversarial Delta Indicator Counter Disinformation Agency Support	Mean Baseline Abs Score	3.268	1.
Adversarial Delta Indicator Defense Industrial Reshoring Support	Big Five Neuroticism Mean %	-3.135	1.
Adversarial Delta Indicator Defense Industrial Reshoring Support	Big Five Openness To Experience Mean %	1.083	1.
Adversarial Delta Indicator Defense Industrial Reshoring Support	Big Five Conscientiousness Mean %	2.016	1.
Adversarial Delta Indicator Defense Industrial Reshoring Support	Sex Female	0.148	4.
Adversarial Delta Indicator Defense Industrial Reshoring Support	Sex Other	1.898	4.
Adversarial Delta Indicator Defense Industrial Reshoring Support	Mean Baseline Abs Score	-3.210	1.
Adversarial Delta Indicator Diplomacy First Preference	Big Five Neuroticism Mean %	0.253	1.
Adversarial Delta Indicator Diplomacy First Preference	Big Five Openness To Experience Mean %	-0.786	1.
Adversarial Delta Indicator Diplomacy First Preference	Big Five Conscientiousness Mean %	0.887	1.
Adversarial Delta Indicator Diplomacy First Preference	Sex Female	-3.140	4.
Adversarial Delta Indicator Diplomacy First Preference	Sex Other	-5.216	4.
Adversarial Delta Indicator Diplomacy First Preference	Mean Baseline Abs Score	-1.017	1.
Adversarial Delta Indicator Integration Funding Support	Big Five Neuroticism Mean %	-2.049	1.
Adversarial Delta Indicator Integration Funding Support	Big Five Openness To Experience Mean %	-2.342	1.
Adversarial Delta Indicator Integration Funding Support	Big Five Conscientiousness Mean %	-3.304	1.
Adversarial Delta Indicator Integration Funding Support	Sex Female	0.073	4.
Adversarial Delta Indicator Integration Funding Support	Sex Other	-3.159	4.
Adversarial Delta Indicator Integration Funding Support	Mean Baseline Abs Score	0.941	1.
Adversarial Delta Indicator Overseas Development Aid Support	Big Five Neuroticism Mean %	1.832	1.
Adversarial Delta Indicator Overseas Development Aid Support	Big Five Openness To Experience Mean %	-0.086	1.
Adversarial Delta Indicator Overseas Development Aid Support	Big Five Conscientiousness Mean %	-2.754	1.
Adversarial Delta Indicator Overseas Development Aid Support	Sex Female	6.153	4.
Adversarial Delta Indicator Overseas Development Aid Support	Sex Other	0.626	4.
Adversarial Delta Indicator Overseas Development Aid Support	Mean Baseline Abs Score	0.113	1.
Adversarial Delta Indicator Trust In Mainstream Journalism	Big Five Neuroticism Mean %	-2.210	1.
Adversarial Delta Indicator Trust In Mainstream Journalism	Big Five Openness To Experience Mean %	2.603	1.
Adversarial Delta Indicator Trust In Mainstream Journalism	Big Five Conscientiousness Mean %	-2.986	1.
Adversarial Delta Indicator Trust In Mainstream Journalism	Sex Female	2.636	4.
Adversarial Delta Indicator Trust In Mainstream Journalism	Sex Other	0.371	3.
Adversarial Delta Indicator Trust In Mainstream Journalism	Mean Baseline Abs Score	0.748	1.

Note. Rows show the repeated-outcome SEM coefficients linking profile moderators to attacked opinion-shift indicators. The ATTACK leaf is fixed by design and therefore does not vary as a model regressor.

Hung, T.-C. and Hung, T.-W. (2022). How china's cognitive warfare works: A frontline perspective of taiwan's anti-disinformation wars. *Journal of Global Security Studies*, 7(4):ogac016. doi:10.1093/jogss/ogac016.

Johns Hopkins University Applied Physics Laboratory and NATO Allied Command Transformation (2021). Cognitive warfare: An attack on truth and thought. Technical report, Johns Hopkins University APL. Sponsored report prepared for NATO Allied Command Transformation.

Kozyreva, A., Lewandowsky, S., and Hertwig, R. (2020). Citizens versus the internet: Confronting digital challenges with cognitive tools. *Psychological Science in the Public Interest*, 21(3):103–156. doi:10.1177/1529100620946707.

Lazer, D. M. J., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., Metzger, M. J., Nyhan, B., Pennycook, G., Rothschild, D., Schudson, M., Sloman, S. A., Sunstein, C. R., Thorson, E. A., Watts, D. J., and Zittrain, J. L. (2018). The science of fake news. *Science*, 359(6380):1094–1096. doi:10.1126/science.aao2998.

Lewandowsky, S., Ecker, U. K. H., and Cook, J. (2017). Beyond misinformation: Understanding and coping with the post-truth era. *Journal of Applied Research in Memory and Cognition*, 6(4):353–369. doi:10.1016/j.jarmac.2017.07.008.

MacKinnon, D. P. (2008). *Introduction to Statistical Mediation Analysis*. Lawrence Erlbaum Associates, Mahwah, NJ.

Matz, S. C., Kosinski, M., Nave, G., and Stillwell, D. J. (2017). Psychological targeting as an effective approach to digital mass persuasion. *Proceedings of the National Academy of Sciences*, 114(48):12714–12719. doi:10.1073/pnas.1710966114.

McCrae, R. R. and Costa, P. T. (1997). Personality trait structure as a human universal. *American Psychologist*, 52(5):509–516. doi:10.1037/0003-066X.52.5.509.

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., and Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*, pages 1–22. doi:10.1145/3586183.3606763.

Paulauskas, K. (2024). Cognitive warfare: The advent of a concept and its implications for nato. Technical report, NATO Strategic Communications Centre of Excellence, Riga.

Roozenbeek, J., van der Linden, S., Goldberg, B., Rathje, S., and Lewandowsky, S. (2022). Psychological inoculation improves resilience against misinformation on social media. *Science Advances*, 8(34):eabo6254. doi:10.1126/sciadv.abo6254.

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2):1–36. doi:10.18637/jss.v048.i02.

Shrout, P. E. and Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, 86(2):420–428. doi:10.1037/0033-2909.86.2.420.